EZWipe vs Redkey USB: Feature & Design Comparison

Oct 21, 2026 · @Sunada Recycling

Summary

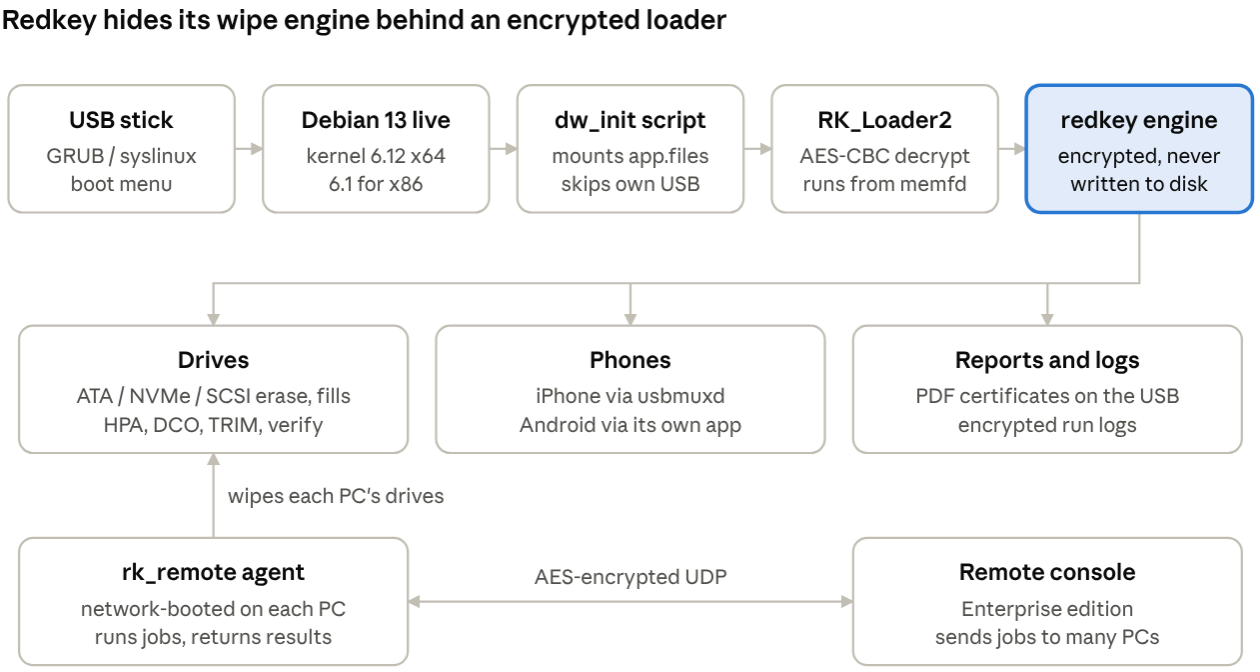
Redkey USB wins on breadth: it boots its own Linux, so it can wipe the system disk, any drive type and phones, with no OS limits. EZWipe wins on trust: fail-closed safety, evidence-based results and a hash-chained audit trail that Redkey does not show.

The biggest gap for EZWipe is reach. Windows 11 blocks the system disk and may block NVMe Sanitize (error 0x13D outside WinPE). A bootable edition (WinPE or Linux) using the same engine closes most of Redkey's lead.

How this was studied: Redkey USB Ultimate v5.74 (hardware rev V5.0), from its own USB stick. Only unencrypted parts were read: boot files, config, language strings, the remote client, and saved reports and logs. The main wipe engine is encrypted and was not decrypted, so its internals are inferred from its UI text and reports, not its code.

Redkey architecture

Redkey is a whole Linux system on a stick, and its wipe engine only ever exists decrypted in memory.



Redkey USB architecture · boot chain, what the engine drives, remote mode

The loader builds a password at runtime, turns it into an AES key with SHA-512, decrypts about 2.7 MB of engine and runs it from memory. Settings live in `config.txt` on the stick; reports and logs are written back to it. Editions (Home, Pro, Ultimate, Enterprise) unlock features; the remote agent and console are the Enterprise tier.

Feature comparison

Redkey covers far more devices and methods today; EZWipe is ahead on safety checks, result honesty and audit integrity. EZWipe column = Phase 1 code in `ezwipe-phase1.zip`.

Area	Redkey USB (v5.74 Ultimate)	EZWipe (Phase 1)	Edge
Platform	Boots its own Debian 13 Linux from USB (x64 + x86)	Windows 11 desktop app (PySide6)	Redkey: no OS in the way
System disk	Wipes it (nothing is running from it)	Always blocked	Redkey
Drive types	SATA HDD/SSD, NVMe, SAS/SCSI, eMMC, USB flash, RAID logical disks	NVMe first; SATA in Phase 3	Redkey

Area	Redkey USB (v5.74 Ultimate)	EZWipe (Phase 1)	Edge
Firmware erase, ATA	Secure Erase, Enhanced, Sanitize Overwrite / Block / Crypto	Catalogued; ATA backend Phase 3	Redkey
Firmware erase, NVMe	Format, Format Crypto, Sanitize, Sanitize Crypto, namespace and controller reset	Catalogued; Windows routes tested in Phase 2	Redkey
Firmware erase, SCSI/SAS	Sanitize Overwrite, Block Erase, Crypto Erase	None	Redkey
Overwrite standards	About 25: DoD 5220.22-M, NIST 800-88 Clear/Purge, IEEE 2883.1, HMG IS5, BSI, NSA 130-1, Gutmann, Schneier, CESG, own RK levels	One single-pass overwrite, marked Clear	Redkey on range
Custom routines	Scripted steps (e.g. ONE, VRQ, ZER, VER) from text files	None	Redkey
Hidden areas	Removes HPA, resets DCO, TRIM	Notes HPA/DCO not covered	Redkey
Frozen drives	Sleep/resume to unfreeze, else falls back to overwrite	Not handled yet	Redkey
Phones	iPhone over USB (`usbmuxd` , `idevicerestore`); Android via its own app	None	Redkey
Remote / fleet	PXE boot client, AES-encrypted UDP link to a console	Sequential batches on one bench	Redkey
Safety	Skips its own USB stick; operator picks disks	Fail-closed checks, typed serial confirm, identity re-check right before command	EZWipe
Result honesty	"Sanitized" even with "No Verification Step"	Completion alone = Inconclusive; needs device status + checks	EZWipe

Area	Redkey USB (v5.74 Ultimate)	EZWipe (Phase 1)	Edge
Verification	Optional: full or 10% read-back	Required for a verified result; per-method checks	EZWipe
Audit trail	Encrypted text logs; report ID per wipe	SQLite, append-only, hash-chained; crash recovery	EZWipe
Reports	PDF: system info, SMART health, temperature, steps used, operator fields, environmental statement, custom logo	JSON + CSV, SHA-256 recorded; PDF in Phase 4	Redkey today
Drive health	SMART hours, wear, temperature graded into levels	NVMe SMART/Health log read	Even
Automation	AutoNuke: wipes on boot with no input	Operator-driven	Redkey
UX extras	20 languages, voice prompts, sound effects, dark mode, serial-console mode	English GUI	Redkey
Editions	Home, Pro, Ultimate, Enterprise	Licensing placeholder (Phase 5)	n/a

Gaps and opportunities

EZWipe's gaps are reach and polish; Redkey's weak spots are trust and evidence, which is exactly where ITAD buyers and auditors push.

What EZWipe lacks today

- No erase on real hardware yet (Phase 2 for NVMe, Phase 3 for SATA).
- Cannot touch the system disk, so a laptop with one drive cannot be wiped from Windows.
- No SAS/SCSI, eMMC, USB flash, RAID or phone support.
- No HPA/DCO handling or frozen-drive recovery.
- No PDF certificate, logo, operator fields or SMART health grading on reports.

- No unattended mode and no network/fleet mode.

Where Redkey is weak (openings for EZWipe)

- **"Sanitized" without proof.** Sample report: Status "Sanitized", Verification "No Verification Step", Verified Sectors 0. EZWipe's rule (completion alone = Inconclusive) is a selling point.
- **Unfinished error handling.** Every report prints "Error Handling: Feature Coming Soon".
- **Vague standard names.** Reports say "Redkey Default Process" next to a firmware erase, with no NIST Clear/Purge mapping. EZWipe ties each method to a category and says when it is provisional.
- **Weak log encryption.** Every log line starts with the same 16 encrypted bytes, which points to a fixed key and IV. Logs are unreadable to the customer and not tamper-evident. EZWipe's hash chain is both readable and tamper-evident.
- **Opaque software.** The engine is encrypted and runs from memory, so a customer or auditor cannot inspect it. EZWipe can offer published method docs and reproducible checks.
- **Under-reported data.** Reports say remapped sector counts "may be under-reported".

Design recommendations

Keep EZWipe's evidence-first core and add reach in this order; each step slots into your existing phases.

1. **Phase 2: prove NVMe on Windows, and plan a boot edition now.** Test the three routes in `docs/technical-study.md`. Whatever fails, a WinPE boot image running the same engine fixes both 0x13D and the system-disk block. Redkey's boot USB shows the market expects this.
2. **Phase 3: ATA with HPA/DCO and frozen-drive handling.** Read HPA/DCO before erase and record it. Reset them and say so in the report. For frozen drives, offer a sleep/resume cycle in the boot edition; on Windows, explain and block.
3. **Phase 4: a PDF certificate that beats Redkey's.** Match its fields (system info, SMART hours/wear/temperature, operator and asset fields, custom logo, unique report ID). Then add what it lacks: NIST category per method, the verdict and evidence level, the hash-chain check and the report's SHA-256.
4. **Verification on by default.** Redkey makes verify optional and still prints "Sanitized". Keep EZWipe's rule, and offer full or sampled read-back with the sample rate printed on the certificate.
5. **Named standards as profiles, not as marketing.** Add DoD 5220.22-M, IEEE 2883.1 Clear/Purge and NIST Clear/Purge as profiles in `policy/`, each mapped to the methods that truly reach that category. Skip Gutmann and other legacy multi-pass

standards unless a customer asks; they add time, not assurance, on modern drives.

6. **Bench automation.** An unattended mode like AutoNuke, but gated: only data disks, only a pre-approved profile, and a printed or saved certificate per drive.
7. **Later: SAS, RAID passthrough, then fleet mode.** A PXE-booted agent reporting to a console is Redkey's Enterprise edge; build it only after single-bench results are solid.
8. **Phones last, if at all.** Redkey's iPhone "wipe" is a single random overwrite (RM1). Apple's own erase is crypto-based. Decide whether this market is worth it before investing.

Clean-room notes

- Everything above comes from Redkey's public behaviour, UI text and its own reports, not from its code. Keep it that way: do not copy its strings, icons, sounds or report layout.
- Build from public specs (NVMe, ACS/ATA, SCSI SBC, NIST SP 800-88 Rev. 2, IEEE 2883) and the open tools Redkey itself ships (`sedutil-cli`, `usbmuxd`, `idevicerestore`), respecting their licenses.